

LocalBuka.com

Production Infrastructure & Enterprise Slack Architecture

This document outlines the ongoing monthly operational expenses required to host, run, and monitor the live **localbuka.com** application. It includes the enterprise-grade event-driven queue architecture designed to handle massive traffic spikes (e.g., 1,000 concurrent actions) without dropping Slack webhook notifications.

Monthly Production Cost Breakdown

Service / Platform	Role in the App	Billing Model	Monthly Cost	Notes / Triggers
Vercel Pro	Frontend UI Hosting	Flat Rate	\$20.00	Per developer seat. Allows commercial use.
Railway (Base)	Platform Access	Flat Rate	\$8.00	Pro plan base fee to prevent service sleeping.
Railway (Usage)	Backend API & DB	Usage-Based	~\$12.00	Estimated vCPU/RAM/Disk consumption.
Sentry (Team)	Error Tracking	Flat Rate	\$29.00	Tracks crashes in real-time.
Upstash QStash	Message Queue	Usage-Based	~\$10.00	Buffers requests to prevent Slack rate limits.
Slack	Webhook Alerts	Free Tier	\$0.00	Receives system alerts.
Total Estimated Cost			~\$79.00 / mo	Varies slightly based on active traffic.

Enterprise Slack Webhook Architecture (No-Compromise)

- **The Message Broker:** To handle massive traffic spikes (e.g., 1,000 simultaneous signups), the system will use an event-driven queue like **Upstash QStash** or **Inngest**. This guarantees zero dropped notifications.
- **Rate Limiting & Retries:** Vercel will instantly dump payloads into the Queue. The Queue is strictly configured to trickle payloads to the Slack Webhook at **1 message per second** to comply with Slack's API rate limits. Failed deliveries will automatically trigger exponential backoff retries.
- **One-Time Engineering Cost:** Implementing a dedicated Message Broker is a complex integration. Estimated labor cost for backend development and testing is **~\$300 - \$500**.